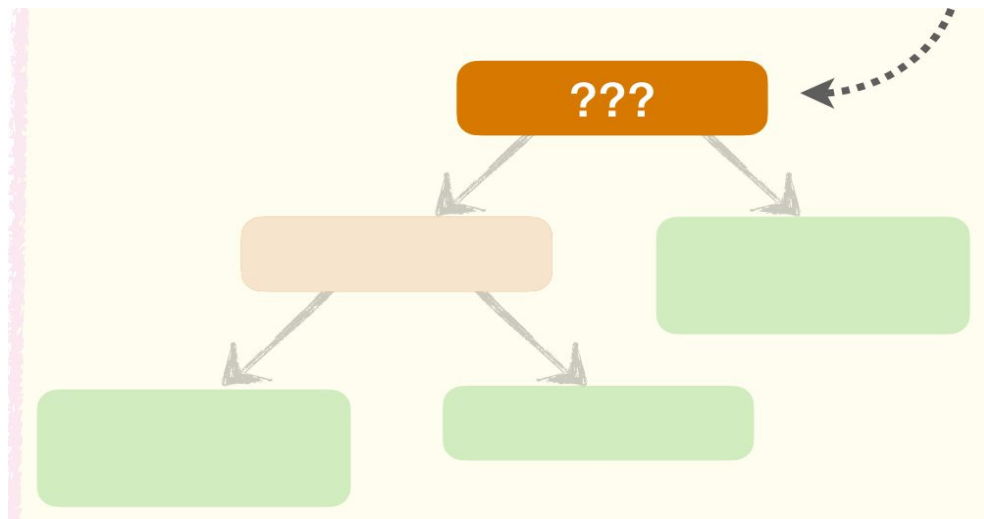
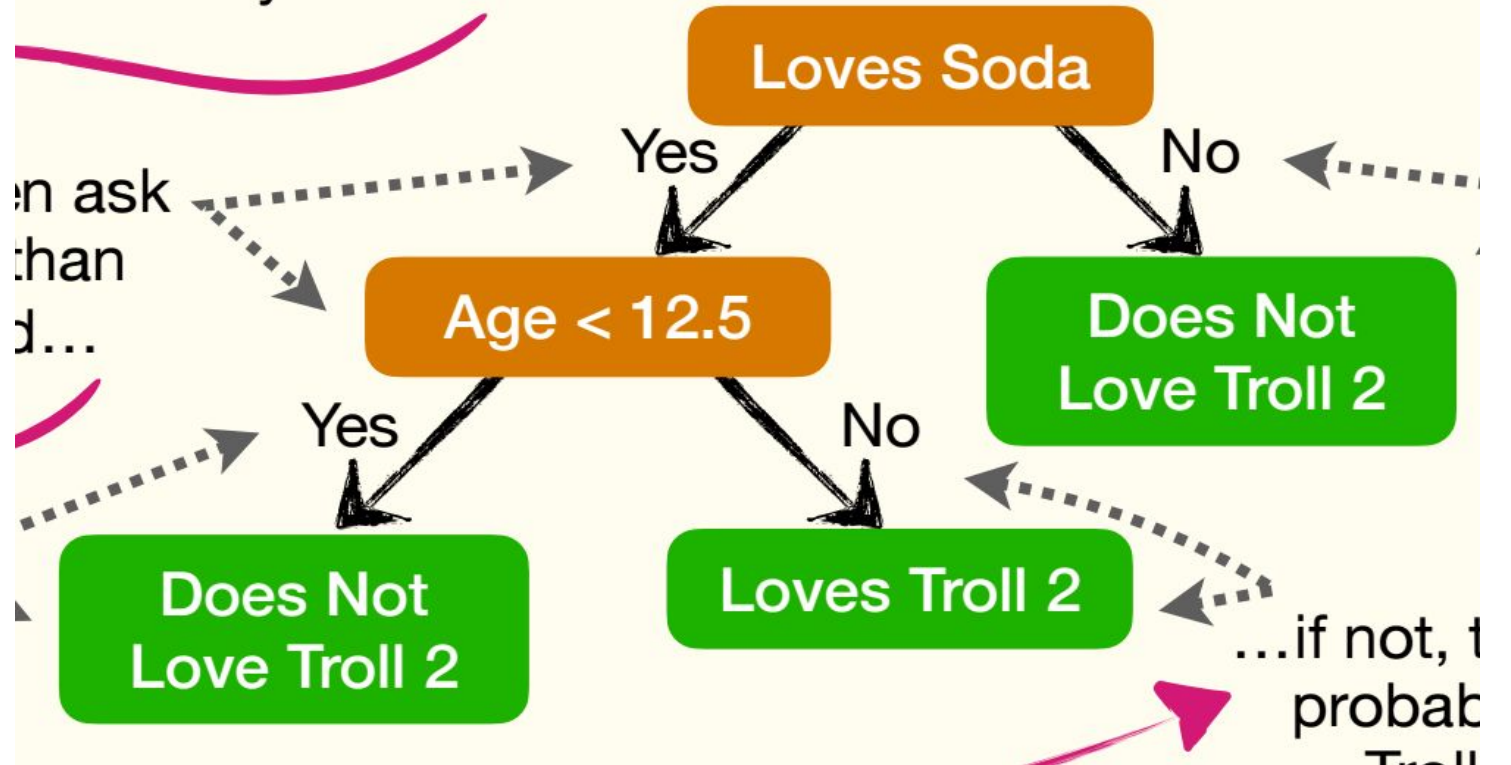


Loves Popcorn	Loves Soda	Age	Loves Troll 2
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

ask if they love Soda...



Loves Popcorn

Yes

No

Loves Troll 2

Yes

No

1

3

Loves Troll 2

Yes

No

2

1

Loves Soda

Yes

No

Loves Troll 2

Yes

No

3

1

Loves Troll 2

Yes

No

0

3

Gini

Impurity = $1 - (\text{the probability of "yes"})^2 - (\text{the probability of "no"})^2$
for a **Leaf**

$$= 1 - \left(\frac{\text{The number for Yes}}{\text{The total for the Leaf}} \right)^2 - \left(\frac{\text{The number for No}}{\text{The total for the Leaf}} \right)^2$$

$$= 1 - \left(\frac{1}{1 + 3} \right)^2 - \left(\frac{3}{1 + 3} \right)^2 = 0.375$$



For the **Leaf** on the right, we get **0.444**.

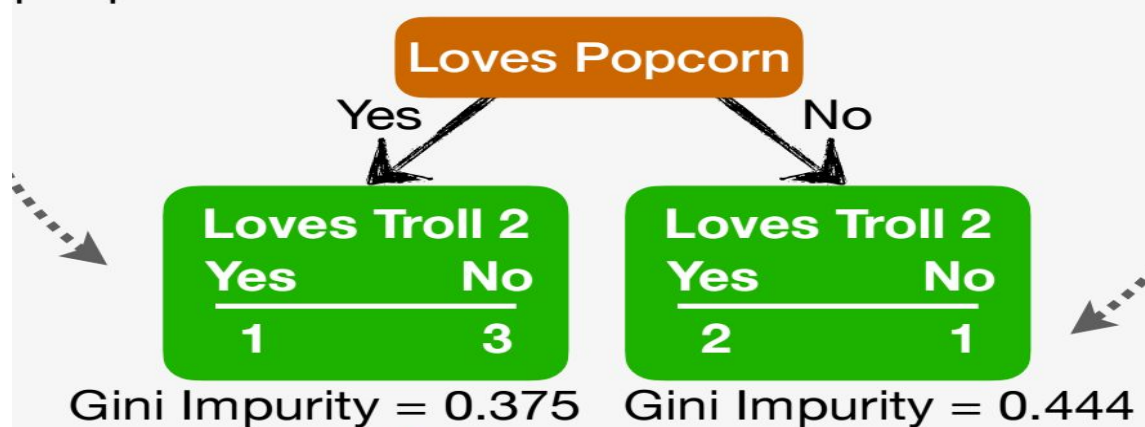


Gini Impurity = $1 - (\text{the probability of "yes"})^2 - (\text{the probability of "no"})^2$ for a **Leaf**

$$= 1 - \left(\frac{2}{2+1}\right)^2 - \left(\frac{1}{2+1}\right)^2 = 0.444$$

On the left has 4 people in it...

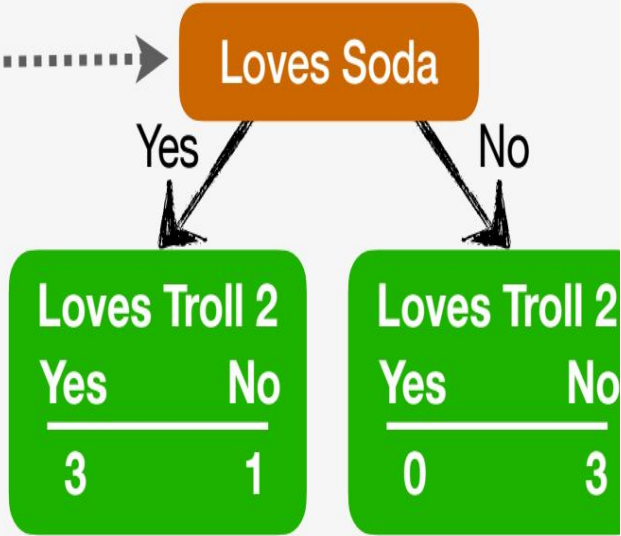
has 3, the **Leaf** the same



$$\text{Total Gini Impurity} = \left(\frac{4}{4+3}\right) 0.375 + \left(\frac{3}{4+3}\right) 0.444 = 0.405$$

ne steps
npurity
214.

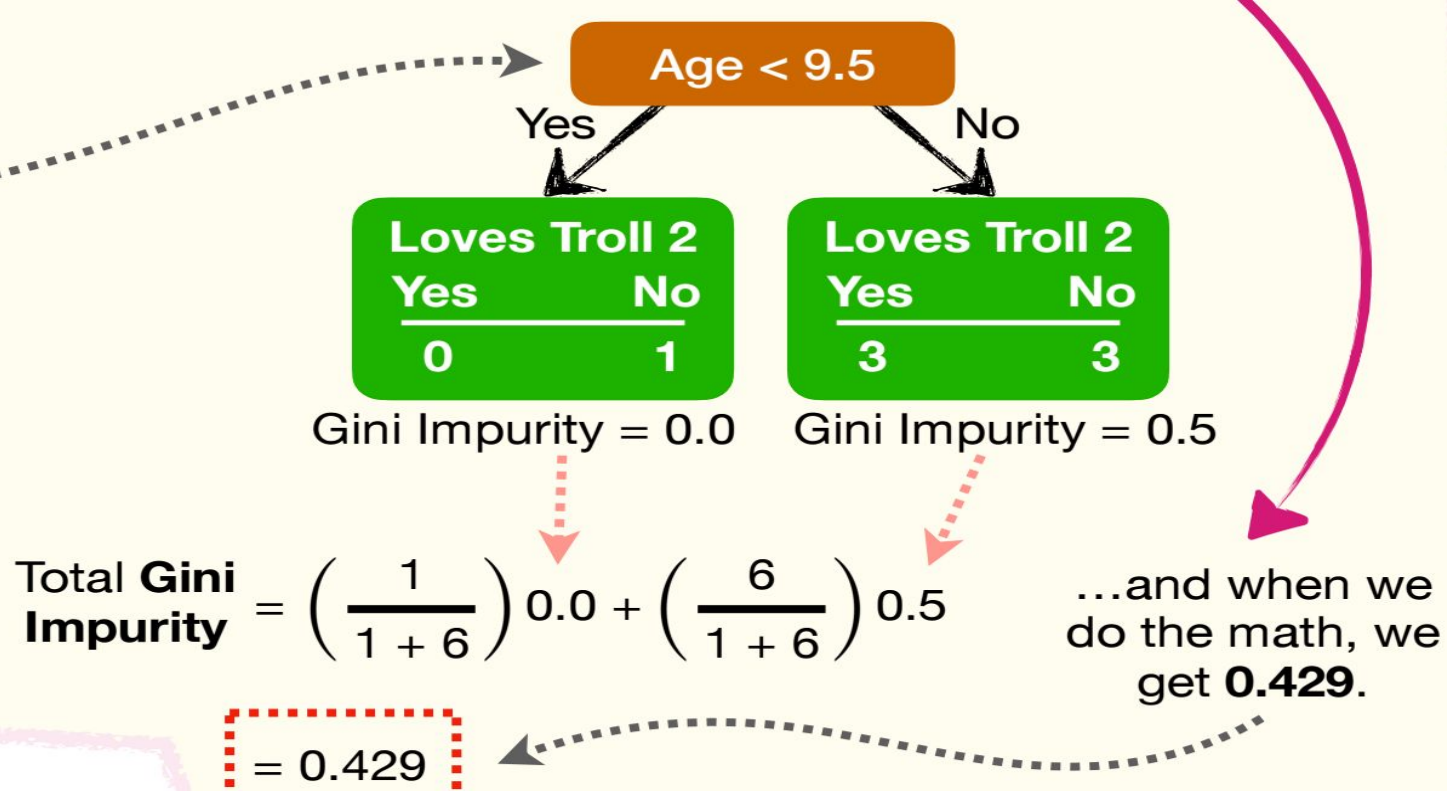
Gini Impurity for
Loves Soda = 0.214



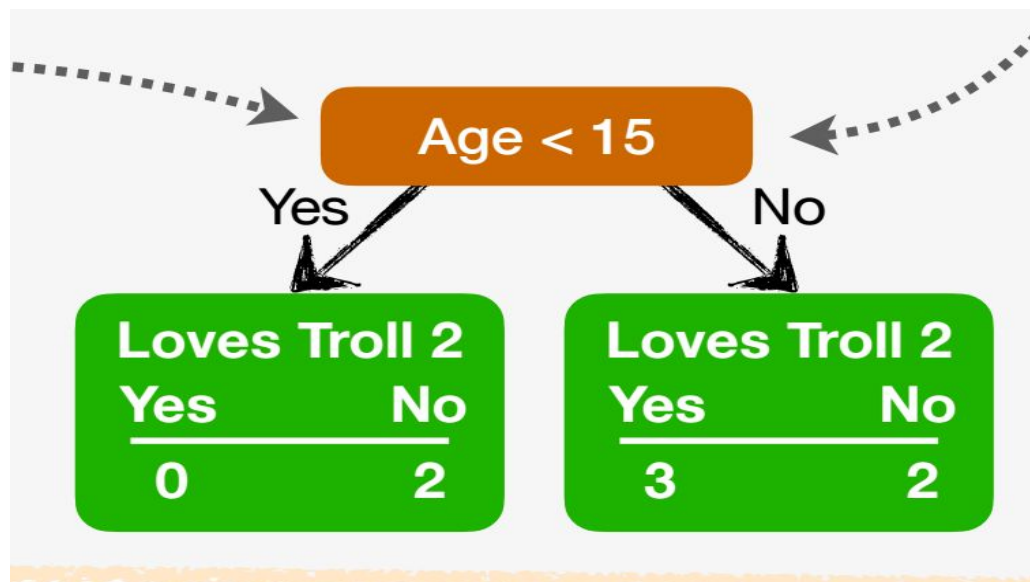
Loves Popcorn	Loves Soda	Age	Loves Troll 2
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

Diagram showing a table with 'Age' and 'Loves Troll 2' columns. To the left of the table, a series of boxes represent potential split points: 9.5, 15, 26.5, 36.5, 44, and 66.5. Arrows point from these boxes to the 'Age' column of the table, indicating the thresholds used for splitting the data.

Age	Loves Troll 2
7	No
12	No
18	Yes
35	Yes
38	Yes
50	No
83	No



Age	Loves Troll 2	Gini Impurity
7	No	Gini Impurity = 0.429
12	No	Gini Impurity = 0.343
15	Yes	Gini Impurity = 0.476
18	Yes	Gini Impurity = 0.476
35	Yes	Gini Impurity = 0.343
38	Yes	Gini Impurity = 0.429
44	No	Gini Impurity = 0.429
50	No	
83	No	



...so we calculated the **Gini Impurities** for each feature...

Gini Impurity for Loves Popcorn = 0.405

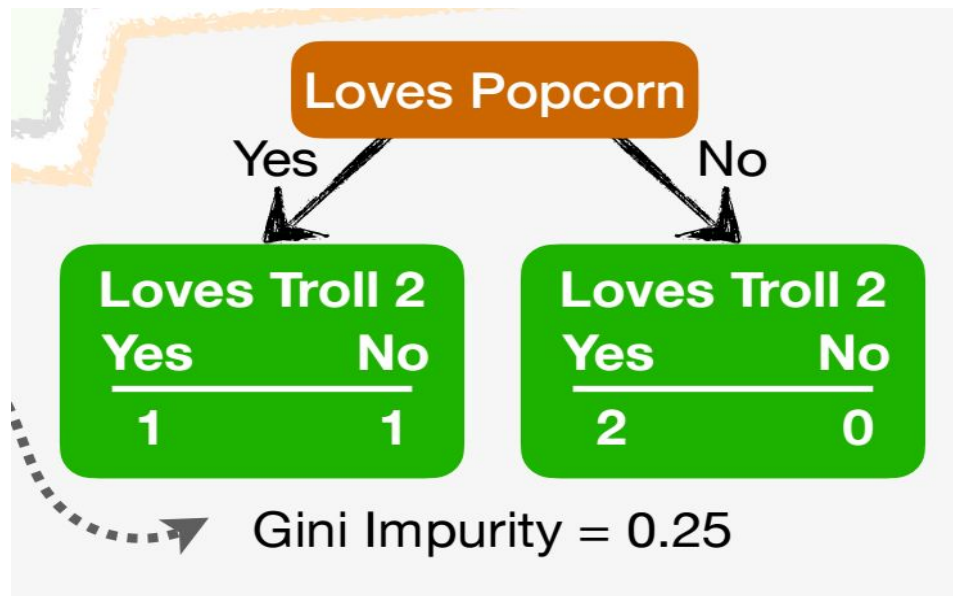
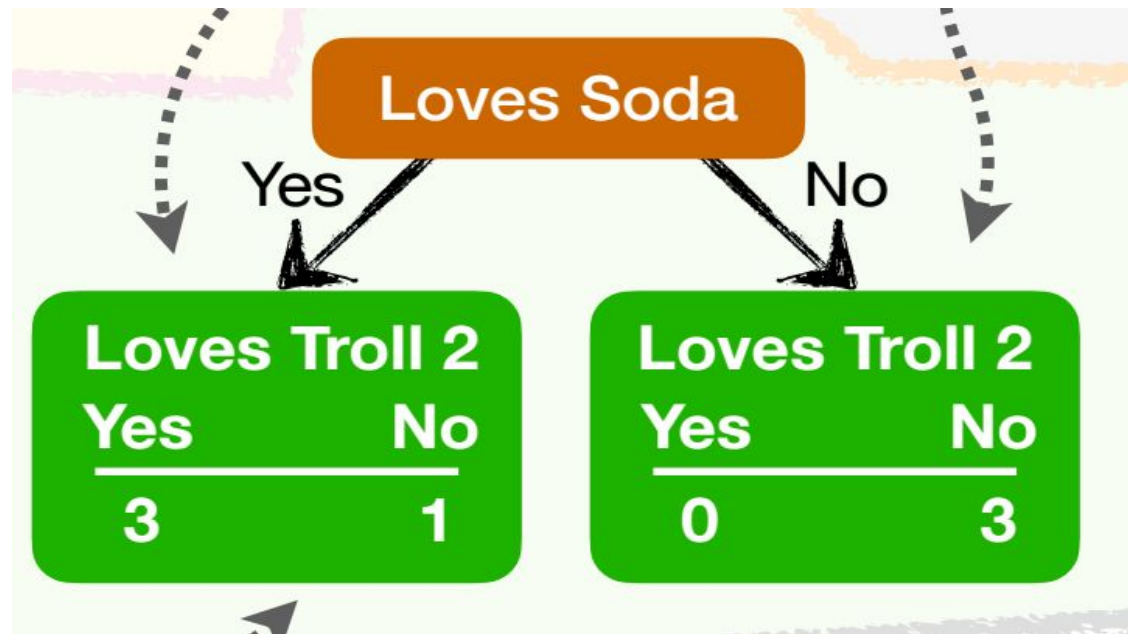


Gini Impurity for Loves Soda = 0.214



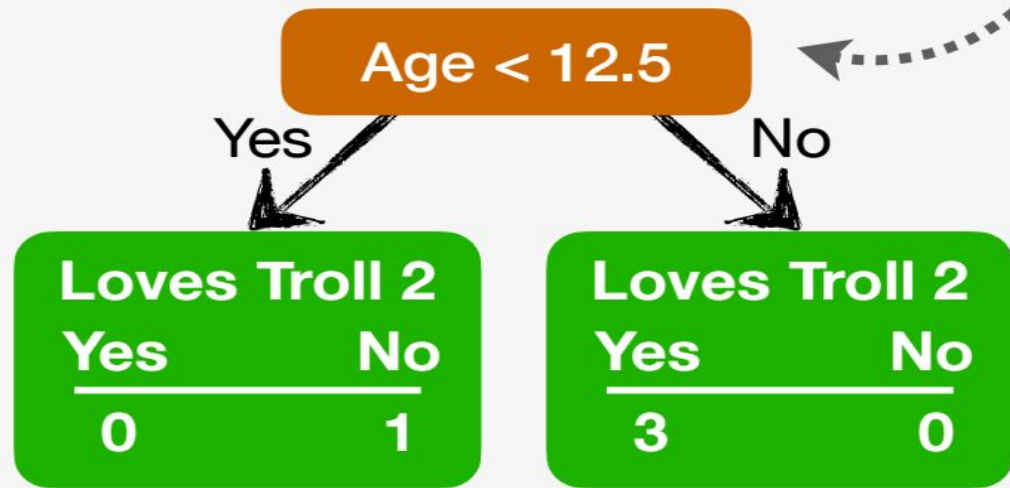
Gini Impurity for Age < 15 = 0.343



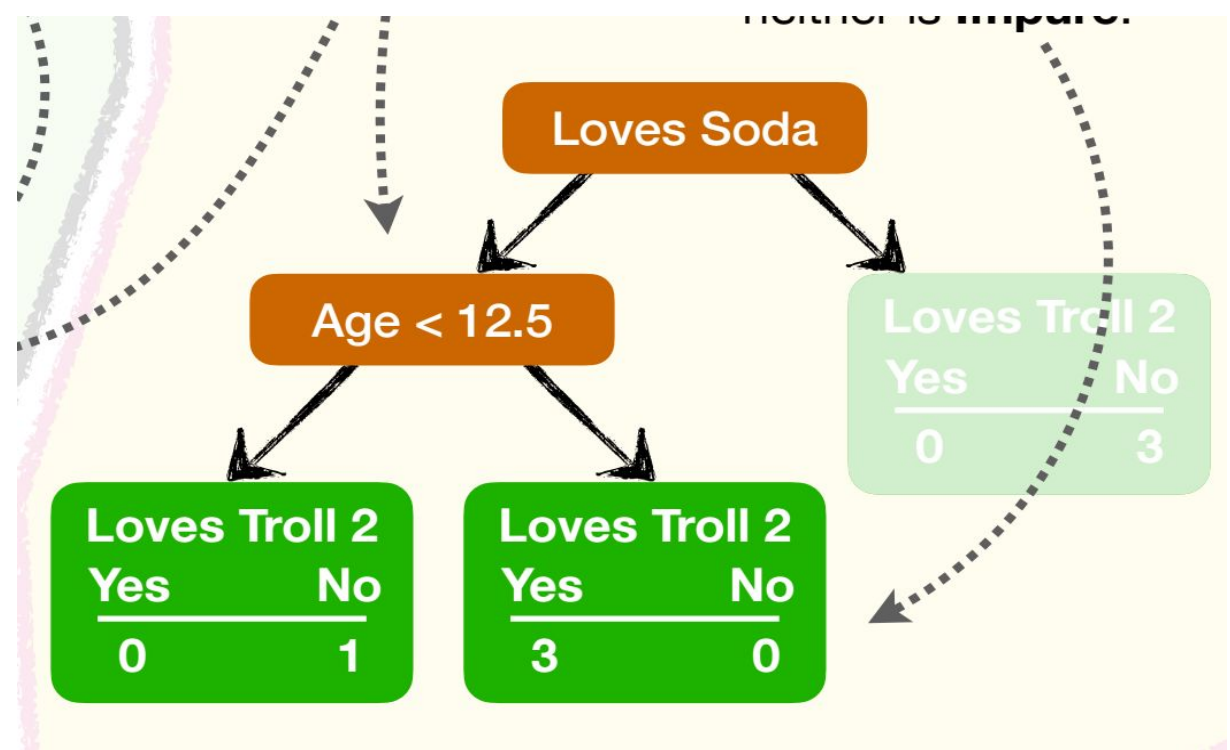


Loves Popcorn	Loves Soda	Age	Loves Troll 2
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

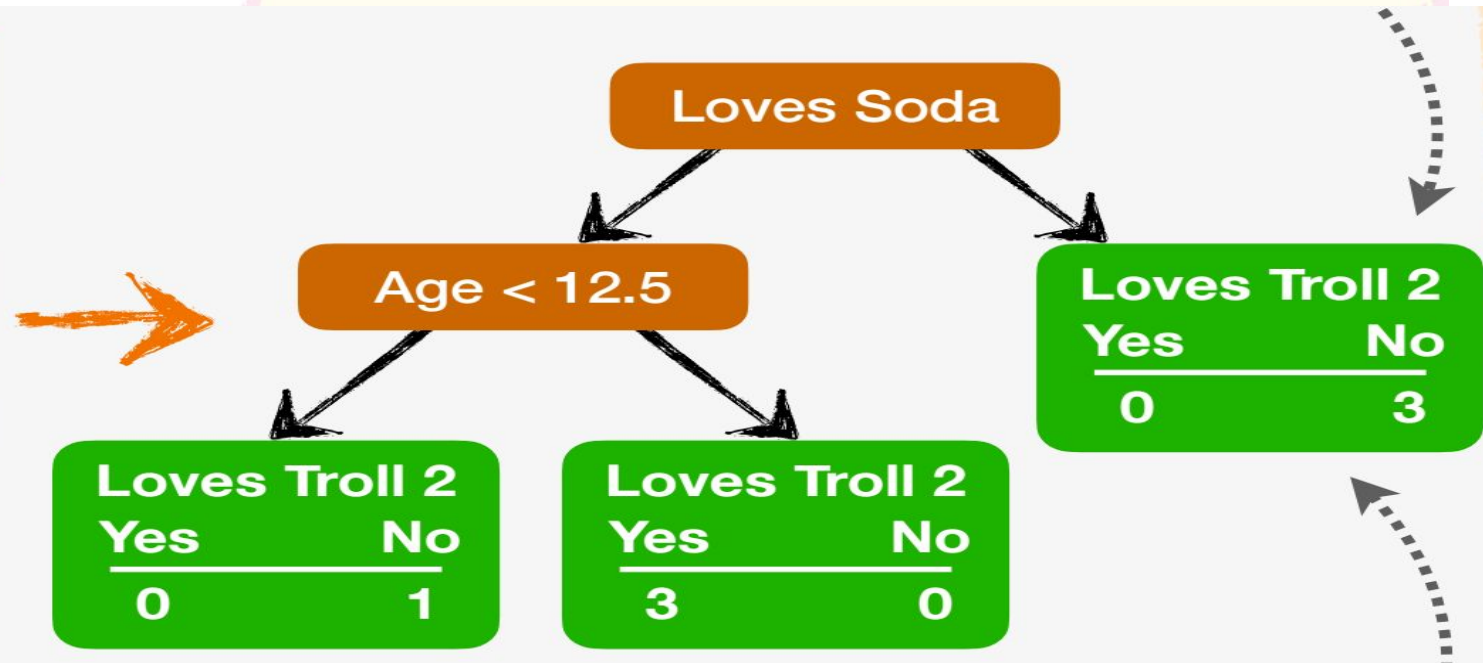
Age values for "Loves Soda" = Yes: 12.5, 26.5, 36.5



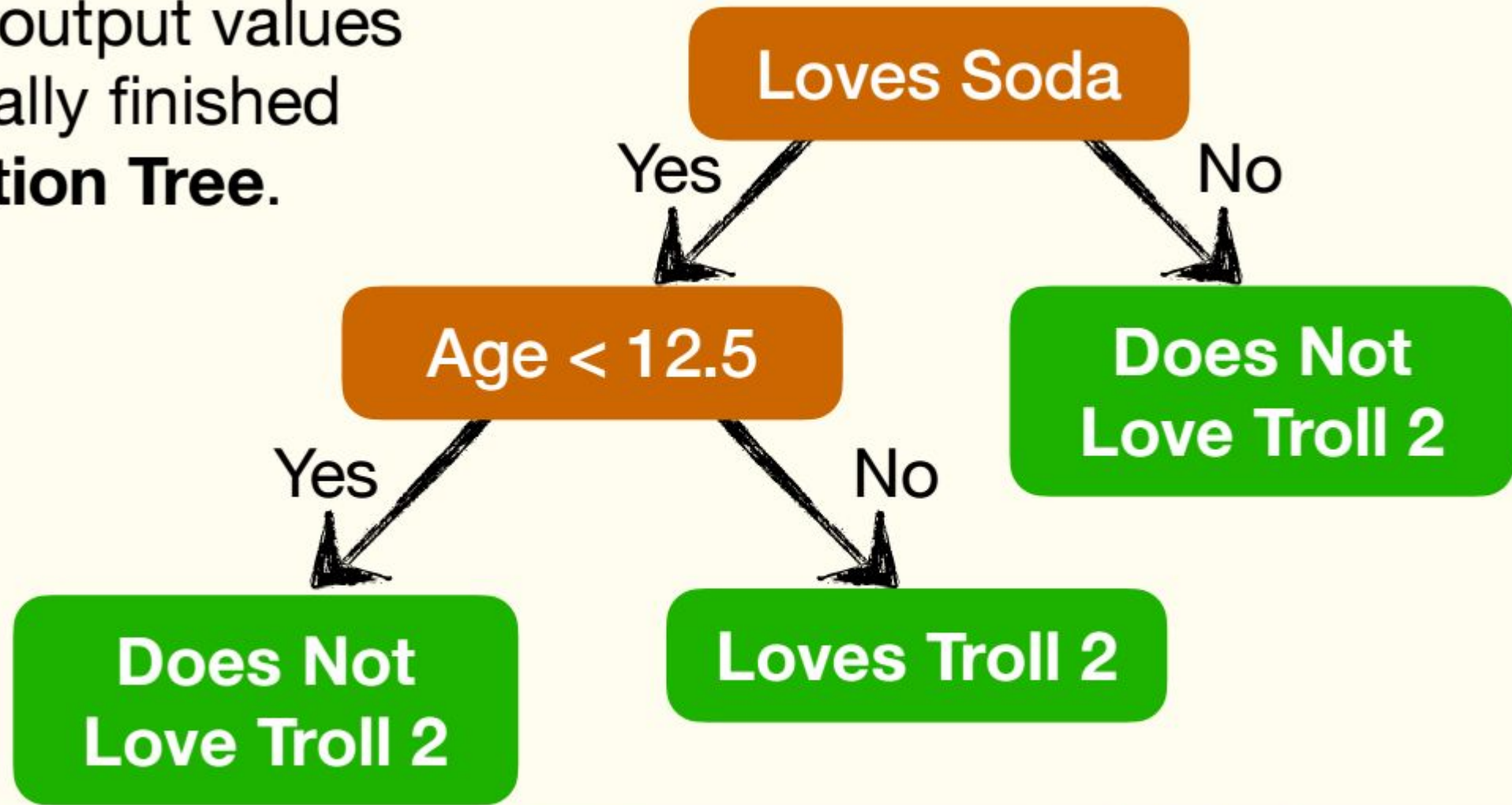
Gini Impurity = 0.0



Loves Popcorn	Loves Soda	Age	Loves Troll 2
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No



ng output values
finally finished
Decision Tree.

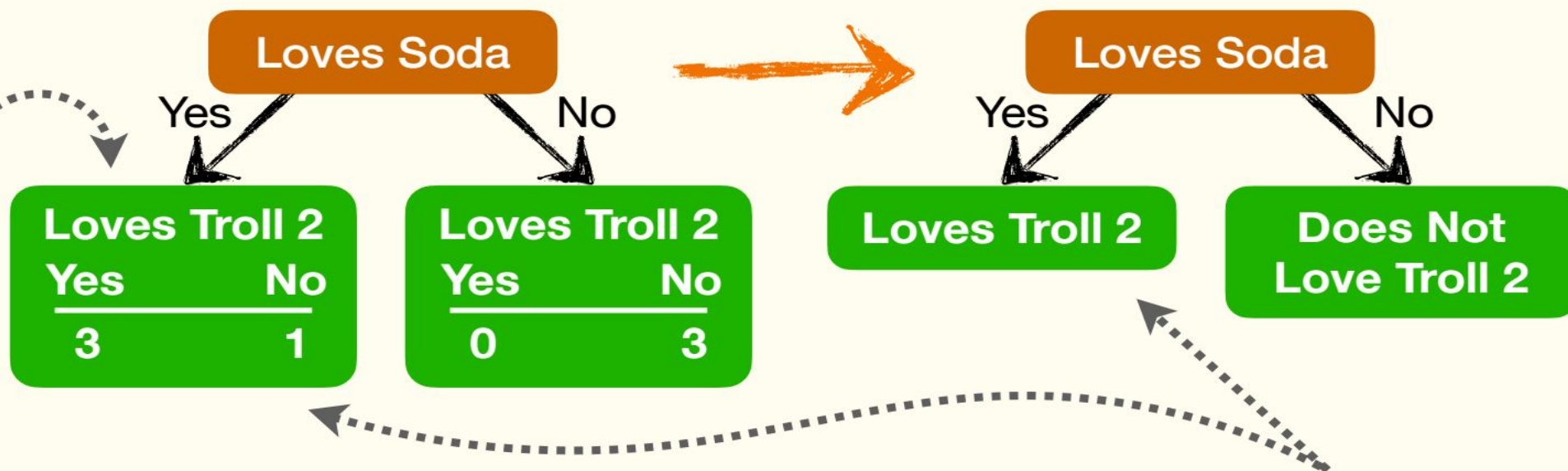


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Alternatively, we can put limits on how trees grow, for example, by requiring **3** or more people per **Leaf**.

If we did that with our **Training Data**, we would end up with this tree, and this **Leaf** would be **Impure**...

...but we would also have a better sense of the accuracy of our prediction because we know that only **75%** of the people in the **Leaf** love Troll 2.



NOTE: When we build a tree, we don't know in advance if it's better to require **3** people per **Leaf** or some other number, so we try a bunch, use **Cross Validation**, and pick the number that works best.

ALSO NOTE: Even though this **Leaf** is **Impure**, it still needs an output value, and because most of the people in this **Leaf** love Troll 2, that will be the output value.